MIT GLOBAL SCALE NETWORK

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MZ 510 Fall 2018

Quiz #1

A lumber yard stocks 1cm diameter wooden dowels in a standard width of 1 meter (standard sized dowels). A customer comes into the yard seeking seven dowels of width 45cm, thirteen of width 37cm, and eight dowels of width 24cm. One possible pattern (there are a number of possible patterns) is to cut two 45cm dowels, leaving 10 cm "waste". The company wants to cut the customer's order so as to minimize the number of standard sized dowels used.

a) Find the other "5 possible patterns" (alternative ways of cutting the standard dowels to produce what the customer needs) in addition to the example given above

b) Formulate a linear program (clearly define decision variables, objective function, constraints) to achieve the above objective (minimizing waste) and meeting customer demand

c) How would you change your formulation if you had to minimize the "total waste".

a)

Pattern	45 cm	37cm	24 cm	Waste (cm)
1	2	0	0	10
2	1	1	0	18
3	1	0	2	7
4	0	2	1	2
5	0	1	2	15
6	0	0	4	4

b) Decision variables:

X : # of standard size dowels used.

P_i : # of patterns of type i cut $(i = 1, \ldots, 6)$

Minimize X

s.t. $\sum_{i=1}^{6} P_i = X$

$2P_1 + P_2 + P_3 \geq 7$

$P_2 + 2P_4 + P_5 \geq 13$

$2P_3 + P_4 + 2P_5 + 4P_6 \geq 8$

$X, P_1, P_2, \ldots, P_6 \geq 0.$

optimal solution:

$X^* = 11$

$P_1^* = 3$

$P_2^* = 1$

$P_3^* = 0$

$P_4^* = 5$

$P_5^* = 2$

$P_6^* = 0.$

c) "Total required" (988 cm = 7×45 + 13×37 + 8×24) is fixed. "Total waste" is "total used" (100X) minus "total required (988). Therefore, it is a function of X, and no change in the formulation is required.

An alternative way to formulate is on the back page.

ALTERNATIVE FORMULATION:

Total waste $= 10P_1 + 18P_2 + 7P_3 + 2P_4 + 15P_5 + 4P_6$.

However, we must also penalize any customer lengths which are made in excess of the customer's order.

S_1: # of dowels of 45cm made in excess of 7 units.

S_2: " " " " 37 " " " " " 13 " .

S_3: " " " " 24 " " " " " 8 " .

Penalty: $45S_1 + 37.S_2 + 24S_3$.

Formulation:

$$\text{minimize } Z = 10P_1 + 18P_2 + 7P_3 + 2P_4 + 15P_5 + 4P_6 + 45S_1 + 37S_2 + 24S_3$$

s.t.

$$2P_1 + P_2 + P_3 - S_1 = 7$$
$$P_2 + 2P_4 + P_5 - S_2 = 13$$
$$2P_3 + P_4 + 2P_5 + 4P_6 - S_3 = 8$$
$$P_1, \ldots, P_6, S_1, S_2, S_3 : \text{non-negative.}$$

* This would give the same solution as one gets 100.X when the above objective function is simplified.